Paper Prototype Evaluation Report

Mobile Application: CoHabit | Human-Computer Interaction 2026/27

Group 11

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1. Introduction and Objectives

This report describes the development, application and evaluation of the first paper prototype of the CoHabit mobile app, as part of the Human-Computer Interaction course (2026/27). CoHabit is a conceptual platform designed to simplify life in shared homes (students, young professionals or families) by centralising the organisation of household chores, the splitting of shared expenses and the scheduling of events and notices in the home.

The main goal of this phase was to create a throw-away paper prototype capable of supporting the essential tasks described in the usage scenarios, and to put it through formal tests with users (classmates from the lab session of 26/28 October). Through these tests, we sought to identify usability flaws and navigation ambiguities, and to gather qualitative feedback in order to iterate on the interface design.

2. User Briefing

The verbal briefing given to each participant at the start of the test session is presented below. This text introduces the purpose of the app and places the user in the problem domain, without revealing how to operate the interface.

> *"CoHabit is a mobile app for people who share a home, such as students or flatmates. It helps everyone in the house stay organised and on the same page: who does which household chores, which shared expenses have been paid and who owes what, and what is happening in the house during the month. Imagine you are João, and you live in a shared flat in Lisbon, 'Casa Santos', with three other flatmates: Tomás, Ana and Pedro. Please interact with the paper prototype as if it were a real app. Touch what you would tap, and think aloud as you go. We are testing the app, not you, so there are no wrong answers."*

3. Test Scenarios

To thoroughly evaluate CoHabit's features, three realistic usage scenarios were defined, each covering different user profiles and use cases within the household ecosystem:

Scenario 1: Four students move in together (Household Management & Shopping/Expenses)

Description: You are Inês, a 20-year-old student who just moved into a rented flat in Setúbal with three friends: Tomás, Beatriz and Diogo. Create a household for the flat called 'Flat 3B' and invite your three housemates. Diogo says the invitation link doesn't open on his phone, so give him another way to join.

The next day, you are Beatriz at the supermarket. Check what the household needs and mark the items as you put them in your basket. At checkout you pay €32.40. Record it so the cost is split equally among the four of you, and check how much each person owes you.

Scenario 2: A couple divides chores and plans a dinner (Chores & Events)

Description: You are Rui, 29, living with your partner Ana. You both feel Ana does most of the cleaning. Set up five weekly chores (bathroom, kitchen floor, laundry, bins and groceries) so that you and Ana take turns doing them.

On Friday you decide to host four friends for dinner on Saturday. Plan the dinner and make sure Ana knows about it.

Scenario 3: Family organisation, Wi-Fi and emergencies (Calendar & Contacts/Utilities)

Description: Carlos's birthday is on Saturday. Plan it so the whole family knows. You have also just changed the router. Update the Wi-Fi password so the kids can reconnect.

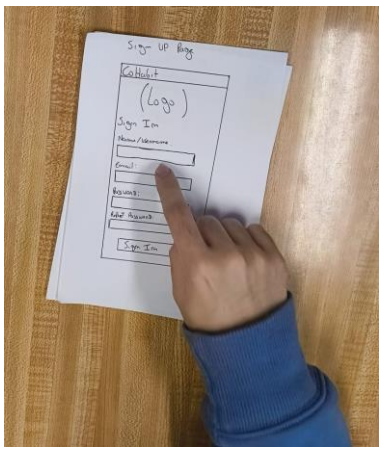
Later in the week, you are Carlos and a pipe starts leaking. Find the plumber's number that Sofia saved, without asking anyone.

4. Interaction Storyboards

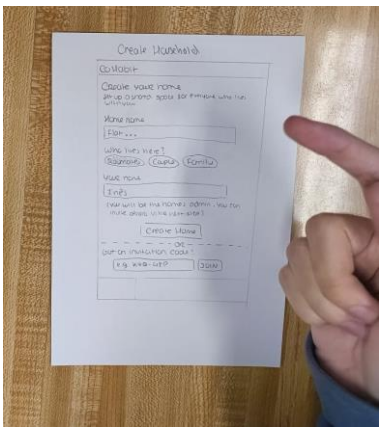
The storyboards below illustrate the sequence of screens and system reactions throughout the execution of the three test scenarios.

Scenario 1

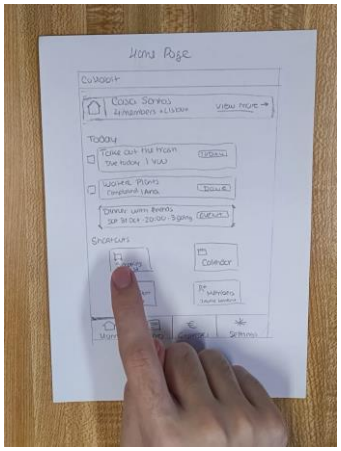
The user starts on the sign-up page (or the login page, if already registered) and fills in their details. They are then redirected to the Household creation page, where they can create a new household or enter an invitation code to join an existing one. The step of inviting the other members via link/code is not illustrated here, but the page exists. Next, the user is taken to the home page ("Home") and taps "Shopping List". The user checks the items the household needs and marks them as purchased. Finally, the user goes to the expenses page and creates a new expense, shared by everyone, splitting the money spent on the shopping.



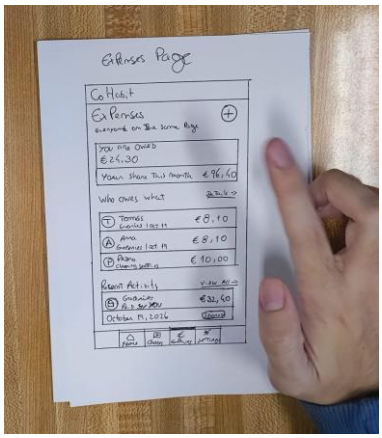
Step 1



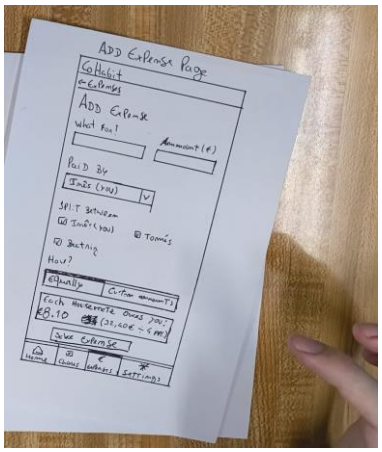
Step 2



Step 3



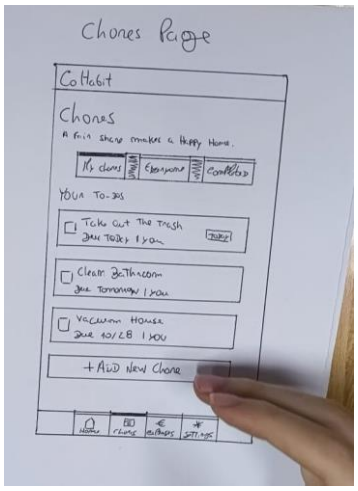
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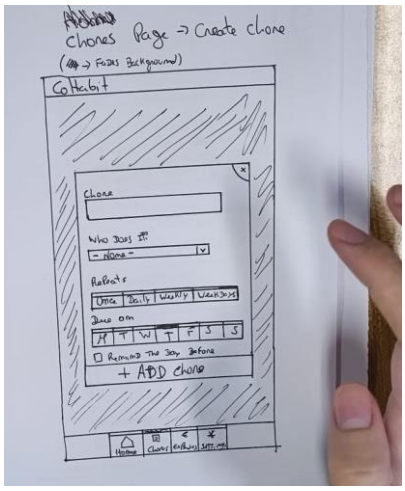
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Scenario 2

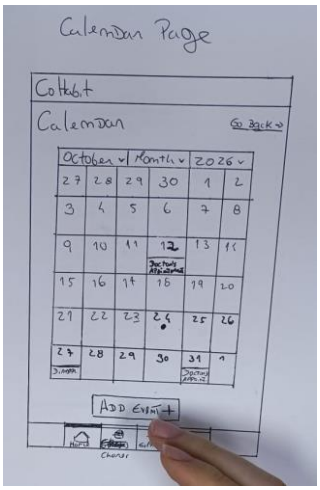
In this scenario, the user starts on the chores page and taps to create a new chore. A small pop-up window with the details of the new chore is displayed, and the user fills in the information and confirms by adding the chore. Then, to schedule the dinner, the user goes to the “Calendar” page (accessible from the “Home” page) and taps “Add event”. They are redirected to the event creation page, fill in the required information and, finally, tap “Create Event”.



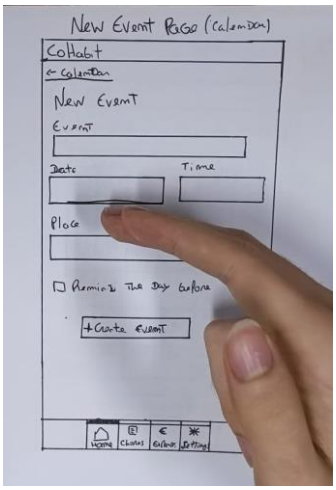
Step 1



Step 2



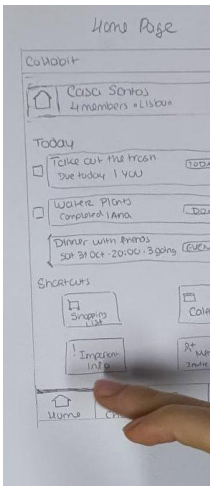
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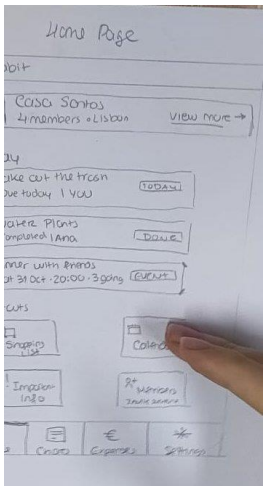
Step 4

Scenario 3

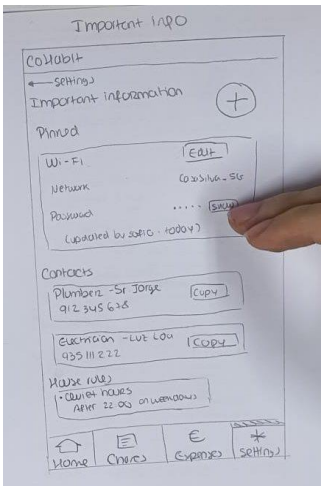
In this scenario, the user again has to access the calendar page to create an event for the birthday. Next, the user returns to the home page and taps “Important Info”, which leads to the Household’s important information page. The user taps the “Edit” button to update the Wi-Fi password. Then, still on the important information page, the user copies the plumber’s number provided in the Household’s contacts.



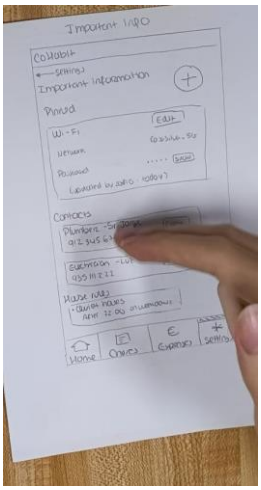
Step 1



Step 2



Step 3



Step 4

5. Observations and Usability Problems

During the hands-on usability testing session, the tests were carried out with several classmates acting as users (Think Aloud technique). The table below summarises the main usability problems identified, along with the respective proposed solutions.

Scenario	Usability Problem Identified	Proposed Solution / Redesign
Scenario 1	Some confusion arose when interpreting how expenses are split.	Simplify the presentation of expense categories.
Scenario 2	One user had difficulty finding the option to add new chores. Creating events also caused some hesitation during navigation.	Give greater prominence to the “Add new chore” and “Create event” options. Introduce a more intuitive event creation sequence and allow event participants to be selected.
Scenario 3	The option to add information to the household was not explicit enough, and doubts arose about the meaning of the question “What kind?”.	Replace the generic question with recognisable categories, such as Wi-Fi, useful contacts and emergency information, accompanied by examples.

6. Conclusion and Next Steps

The tests carried out indicate that CoHabit has a broadly intuitive and accessible interface. The three participants managed to complete most of the flows without significant difficulty, which suggests that the overall organisation and the main elements of the app are understandable. The hesitations recorded were isolated and were mainly related to the clarity of some instructions, the discoverability of certain actions and the interpretation of how expenses are split.

As next steps, we propose clarifying the household creation process and the invitation methods, making the split of expenses more explicit, giving greater prominence to creating chores and events, and reorganising the useful information into clear categories. Selecting event participants should also be explored, and the navigation revised to reduce unnecessary returns to the home page.